

MERGE4WARD STRATEGY MEMO · PHASE 2 EVOLUTION

From Manufacturer to Platform —
The ZÜBLIN Model as Our Year 2-5 Roadmap

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FROM:	David C. Ainsworth, AIA — Predevelopment lead, Merge4Ward partner
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RE:	Strategic positioning of Merge4Ward beyond Phase 1 — what the European model teaches us about where this should go
STATUS:	Internal partnership discussion document. Not for external distribution.

This memo proposes a strategic refinement to the Merge4Ward business plan we have been working from since the October 2025 strategy document. Phase 1 — manufacturing LLC, Mid-Atlantic CLT/SIP supply, M4D as anchor customer, LEEP DE as workforce-training partner — does not change. What changes is what we point Phase 1 toward. The October plan implicitly aimed Phase 1 at becoming a larger version of itself: a regional commodity panel manufacturer. The proposal here is that we aim Phase 1 at becoming something materially different — a vertically integrated timber project delivery platform — because the platform structure builds substantially more durable equity for us as owners, creates substantially more and better jobs for the workforce we are training, and leaves the legacy-structure question genuinely open for a partnership conversation we have not yet had.

The case rests on a single observation about a European firm called ZÜBLIN Timber. Reading their business carefully changed how I think about what Merge4Ward should be when it grows up. The short version is below; the implications for our Phase 1 decisions, and for the wealth-and-jobs argument that justifies those decisions, are the part that matters.

BOTTOM LINE

ZÜBLIN Timber is not a CLT factory. It is a vertically integrated timber project delivery platform with manufacturing as one node among several. They sell technology, design support, production, delivery, connectors, and signature capabilities — and the integration is the moat. The moat is what matters here, because it does two things at once: it builds durable equity for us as owners over twenty years instead of five, and it creates more jobs across more disciplines than a panel factory ever could. Most US CLT producers compete on price-per-cubic-meter at the bottom of the value chain — narrow margins, narrow workforce, narrow legacy. ZÜBLIN sits at the top of the value chain — broader margins, broader workforce, deeper legacy. Merge4Ward should aim there too, in Phase 2 and

beyond. Three Phase 1 decisions need to be made now to keep that path open: site footprint, building layout, and the operating agreement. Each is addressed in Section 4.

1. Who ZÜBLIN Timber Is

ZÜBLIN Timber GmbH is a wholly-owned subsidiary of STRABAG SE, the Austrian construction multinational with about €17B in annual revenue and 73,000 employees — one of the largest construction firms in Europe. ZÜBLIN Timber operates from Aichach, Germany with branch offices in five other German cities. They are part of STRABAG's publicly stated commitment to be climate-neutral by 2040.

They are not the largest CLT producer in Europe. Several other firms (KLH, Stora Enso, Binderholz) outproduce them on raw panel volume. What makes them strategically distinctive is what they sell *around* the panel.

What they actually sell

Pulled directly from their corporate brochure:

- LENO® Cross Laminated Timber — solid CLT panels up to 4.80 m × 20 m (about 16 ft × 65 ft), thicknesses 60–320 mm
- LENO®-ADD — prefab wall panels built on the LENO® CLT base with insulation, cladding, and interior finishes installed at the factory; available unfinished, with substructure, or fully finished including wood façade
- MOLENO timber-hybrid system — extensively prefabricated CLT exterior walls combined with wide-span concrete floors, designed for serial / standardized residential construction at affordable price points
- LENO®-HYBRID — CLT panels with substituted high-stress layers (LVL or OSB instead of solid lumber) for large-span or point-supported conditions
- LENO®-3D — uniaxially or biaxially curved CLT, radii ≥ 9 m, dimensions up to 4.50 m × 14.50 m — a capability no US producer currently offers
- Stressed-skin panels — high-performance ribbed panels for spans up to 20 m
- Laminated veneer lumber components — large-format reglued and pressure-impregnated LVL for façade, columns, and outdoor structural use
- Tools and joining technology — column supports, tension anchors, beam connectors, screw connectors, screw foundations, sold under their own brand

What they do for the customer

Service stack, again straight from their brochure:

- Pre-project consulting and structural pre-analysis
- Construction proposals (the European equivalent of a full-design-to-build offering)
- Structural design and workshop planning

- Element planning — translating architectural and structural drawings into element-level shop drawings, factory-machined to fit
- CNC plus robot manufacturing, with skilled-employee manual trimming and finishing
- On-time delivery with special transports as required
- Project support throughout the construction phase

Their target customer in the brochure is, exactly: "Carpenter, Timber Builder, Architect, Engineer." They do not market to general contractors or owners — they market to design and construction professionals. That is the same customer M4D talks to every week. Their cultural framing throughout the brochure is partnership-coded: "From Person to Person — Personal and Authentic" and "We build on your design and support you in the process."

2. The Strategic Insight

Read that product list and that service list together and a pattern surfaces. ZÜBLIN does not sell CLT panels. They sell project-delivery outcomes — design support, fabrication, finishing, connectors, delivery, and field support — with manufacturing in the middle as one capability among many. The factory is the anchor of the operation, but it is not the product.

Compare this to most US CLT producers — SmartLam, Nordic Structures, Mercer, Element5, Vaagen Brothers, Sterling, Freres. They are commodity panel manufacturers. They sell panels by the cubic meter. The architect, engineer, and contractor figure out everything else from independent sources — connectors from Simpson or Rothoblaas, finishes from cladding subs, design support from independent timber engineers, on-site coordination from the GC. The producer's revenue per project is constrained to the panel margin; the rest of the value chain is captured by other parties. They are stuck at the bottom of the value chain competing on per-panel price.

ZÜBLIN sits at the top of the value chain. Their revenue per project is panel material plus design fees plus connector sales plus finishing margin plus delivery markup plus signature-capability premium where it applies. Per-project revenue is multiples of what a commodity producer earns from the same project, because they are selling the whole timber-construction outcome, not a piece of it.

Why this matters for the five of us — the owner-side argument

The integration is the moat. The moat is what holds the equity value of this company across a twenty-year horizon. A commodity panel manufacturer is a fundamentally cyclical business — it rises with construction demand and falls with it, it competes on price with whoever else has a press, and the equity value of the partnership tracks the construction cycle. Five years of strong margins, two years of squeezed margins, repeat. The exit value at any point depends on what the construction market is doing that year.

A vertically integrated platform does not have that profile. The design-services revenue is countercyclical — when construction slows, more projects need feasibility help and predevelopment cost intelligence, and our design-services line picks up. The connector and tool

distribution business is steady-state — projects need connectors regardless of who manufactures the panels. The signature capabilities (curved CLT at Phase 4, MOLENO-equivalent multi-story at Phase 3) are marquee-project work that earns premium pricing and inbound calls rather than competitive bids. The platform's revenue mix is diversified across cyclical and non-cyclical lines, which means the equity value of the partnership compounds rather than oscillates. Twenty years of compounding equity is a fundamentally different wealth-building outcome than five years of strong margins followed by a sale.

That is the owner-side case for the platform model. Each of us — David, Trevor, Leo, Travis, Gwen — captures the wealth-building benefit of the integration directly through our partnership equity. The moat we build at Phase 2 and beyond is not abstract; it is the literal protection of the equity value we hold.

Why this matters for Wilmington — the workforce argument

A panel factory creates a specific kind of job: production-line operator, CNC technician, panel assembly, dispatch and logistics. These are good jobs. Phase 1's projection of 22-35 manufacturing positions at \$26-\$34/hour by Year 3 stands as the workforce baseline. Those numbers don't change.

What the platform model adds is jobs in adjacent disciplines that do not exist at a commodity panel factory. Design and engineering services creates technical-track positions — drafters, project coordinators, structural and envelope analysts — at \$30-\$45/hour. Connector and tool distribution creates logistics and inside-sales positions at \$24-\$34/hour. Pre-finished prefab wall production creates cladding-installation, paint-booth, and quality-control positions at \$26-\$36/hour. By Phase 3, the same campus that houses a 22-35-position manufacturing operation is also housing another 30-50 positions in design services, distribution, and prefab finishing. The total workforce at Phase 3 is double what the commodity-factory model produces, across a broader range of disciplines, with multiple training tracks running through LEEP DE rather than a single manufacturing track.

That is the workforce-and-community case for the platform model. More jobs, better jobs, broader career pathways, deeper grant defensibility, and a stronger story for OED, Wells Fargo, Enterprise, and every funder we will engage over the next decade.

THE TRANSFERABLE LESSON

What's importable here is not the German manufacturing technology. The technology can be acquired or licensed. What's importable is the strategic architecture — the decision to build a project delivery platform around a manufacturing capability, rather than to build a manufacturing operation that sells material to whoever shows up. That architectural decision is what protects partner equity over twenty years and what doubles the workforce footprint in Wilmington over Phase 2 and 3.

3. What This Means for Merge4Ward — Phase Evolution

Phase 1 doesn't change. We still need to build a manufacturing facility, hire and train workforce, fabricate the first SIP and CLT runs, and deliver Claymont and the next two or three projects to demonstrate execution. The Phase 1 plan in the October 2025 strategy document is right; we just need to recognize that Phase 1 is the foundation of a larger structure, not the structure itself.

What changes is what we point Phase 1 toward. Here is the proposed Phase 2 / Phase 3 / Phase 4-5 evolution:

Phase 2 (Year 2–3) — From manufacturer to platform

Three product line additions, each capturing higher margin than panel sales alone:

- **Pre-finished prefab wall product.** ZÜBLIN's LENO®-ADD parallel — a Merge4Ward standard offering combining CLT or SIP structural panels with factory-installed insulation, cladding, and interior finishes. The Claymont pilot is already a small-scale version of this; we just productize what we are already doing for one project as a recurring SKU. "Merge4Ward Wall — fully-finished, factory-trimmed, delivered ready to install." Significantly higher margin than raw panel sales.
- **Connector and tool distribution.** ZÜBLIN sells their own connectors under a separate brand alongside their panel business. US CLT/SIP projects need connectors regardless of where the panels come from. Merge4Ward can become the regional distributor for Rothoblaas, Simpson Mass Timber, and similar product lines — selling to projects whether they buy our panels or someone else's. Connector margin is 30-40%. This is a parallel revenue line that does not require additional manufacturing capacity to scale, and it lets us build customer relationships with projects we would not otherwise touch.
- **Design and engineering services.** ZÜBLIN sells consulting, structural pre-analysis, construction proposals, and element planning. M4D and Aubyn already have the in-house capability to provide these services. Phase 2 formalizes them as a Merge4Ward fee service — billed separately from panel material — to architects and engineers who buy our panels and to projects exploring CLT/SIP feasibility before they buy panels from anyone. This converts a capability we already have into a revenue line we are not currently capturing.

Phase 3 (Year 3–4) — Multi-story residential, factory finishing

- **MOLENO-equivalent multi-story residential system.** ZÜBLIN's MOLENO concept — extensively prefabricated CLT exterior walls combined with wide-span concrete floors — is the model that solves the structural-grid problem in 4-story affordable housing. Most US affordable housing of this typology is currently wood-framed because concrete floors are expensive and grid-constrained. MOLENO uses less concrete (wide-span floors instead of grid floors) while opening up unit-plan flexibility. For Wilmington Housing Authority's pipeline (Lobdel Avenue and beyond) and for the broader Mid-Atlantic 3-to-4-story residential market, this is a directly applicable system. Phase 3 product: "Merge4Ward Mid-Rise — multi-story residential timber-hybrid system, Mid-Atlantic edition."
- **Move finishing operations factory-side.** By Phase 3, the prefab-wall product line is mature enough to move the exterior finishing — cladding installation, weather barrier, window pre-mounting — from job site to factory. Panels arrive on site nearly complete. This captures the

labor margin from field installation, tightens quality control, and reduces site-time exposure for our customers. Logistics constraint: factory-pre-assembled panels are larger and heavier than bare panels, so transport and crane planning needs evaluation against 53' × 8'-6" flatbed limits.

Phase 4–5 (Year 4–5+) — Signature capabilities

- **Curved CLT capability.** ZÜBLIN's LENO®-3D is a marquee product. No US manufacturer currently produces curved structural CLT. Adding a 3D press at Phase 4 makes Merge4Ward the only US producer of the capability — earning press, awards, and inbound calls from architects nationally. Capital investment is meaningful (\$1.5M+ for the press alone), so this only justifies once standard panel production is at maturity and the marquee-project pipeline supports the equipment. But the strategic position is real and worth aiming at.
- **Legacy structure — a Phase 4-5 partnership conversation worth opening.** By Phase 4, we will have a mature, profitable, multi-line platform. The five of us will need to decide what we want this company to become for the long term — what kind of legacy structure preserves the wealth, protects the workforce, and honors the mission across the next generation. There are several options worth thinking about: continued partner-owned LLC with succession planning and transfer mechanics, conversion of part or all of the operation to a worker cooperative, ESOP structure, or a hybrid that keeps strategic functions partner-owned while opening operational ownership to the workforce. None of these is a decision for now. They are options for a partnership conversation we should plan to have somewhere in Year 3 or Year 4, when we have enough operating history to make the choice with real information rather than abstraction. The Phase 1-2-3 platform structure proposed in this memo leaves all of these options open. The commodity-factory model would foreclose most of them.

4. Three Phase 1 Decisions That Keep This Path Open

This is the part of the memo that needs partnership-level alignment in the near term. The Phase 2 / 3 / 4-5 evolution above is a long-horizon roadmap, but it depends on three decisions we are making in the next six months. If we choose any of the three the wrong way, the larger path is foreclosed. None of these are difficult; they just need to be conscious.

#	Decision	What this means in practice
01	Site footprint — evaluate at Phase 3 ultimate, not Phase 1 immediate.	The current site search prioritizes Phase 1 needs (30,000–45,000 SF building, 1–6 acres of yard, Port-adjacent for raw timber import). Phase 3 needs are larger — finishing operations need additional floor space, cladding storage, paint booth or finishing line, and possibly a curved-CLT press at Phase 4. The total Phase 3 footprint is closer to 8–12 acres, not 1–6. Site A's existing-industrial 4–6 acres is right for Phase 1 and likely accommodates Phase 2-3 if we can secure adjacent expansion parcels. Site B's 1–2 acres is only viable if combined with neighboring expansion. Trevor and Leo: please reframe the site search to evaluate ultimate Phase 3 footprint, not just Phase 1.

02	Building layout — design Phase 1 building for Phase 3 expansion.	Foundation depths, ceiling heights, crane bay dimensions, electrical service, and building shell flexibility should accommodate Phase 2 finishing operations and Phase 3 expansion — not just Phase 1 SIP press and CLT staging. The cost differential between a Phase-1-only building and a Phase-3-ready building is not large at the design stage; the cost of retrofitting a Phase-1-only building at Phase 2 is enormous. Travis: when the architectural design begins, build the program around Phase 3 capacity with phased build-out of equipment, not around Phase 1 alone.
03	Operating agreement — write to accommodate scope expansion.	The Merge4Ward LLC operating agreement, when finalized with the Delaware entity-formation attorney, should explicitly contemplate (a) future product lines beyond CLT/SIP panels, (b) sister-entity formation for related operations under shared partner ownership, and (c) flexibility for whatever long-term legacy structure the partnership eventually chooses — partner-owned LLC with transfer mechanics, worker cooperative, ESOP, or hybrid. Generic boilerplate language will likely foreclose options the business model will eventually want. This is a lawyer-conversation item, not a partnership debate — but the partnership needs to brief the lawyer correctly. Leo: please add scope-expansion and legacy-flexibility language to the operating-agreement drafting checklist before the next attorney session.

5. What This Does Not Ask You to Do

This memo deliberately does not ask the partnership to:

- Reach out to ZÜBLIN now. The credibility weight of an outreach today is light — we have no signed MOU between the five partners, no site, no first production run, and no closed grant. The same email twelve months from now, with a Wells Fargo decision and Claymont units in production, lands very differently. I have drafted an introductory email and am holding it in queue rather than sending it. (One narrow exception: if Riverfront East Parcel B advances with RDC, ZÜBLIN's curved-CLT capability becomes uniquely relevant to a public pavilion concept, and project-specific outreach earlier becomes correct. Trevor — flag if Parcel B advances.)
- Add new product lines to the Phase 1 plan. Phase 1 is still SIP and CLT panel manufacturing only. The Phase 2-5 expansion is a roadmap, not an immediate scope addition. Stay focused on getting one product right before adding scope.
- Change the workforce training program scope. LEEP DE workforce training as planned for Phase 1 is unaffected by this memo. The Phase 2-3 expansion may eventually create additional training tracks (cladding finishing, connector installation, curved-CLT fabrication), but that is a Year 2-3 conversation, not a near-term scope change.
- Foreclose any legacy-structure options. The platform model proposed here keeps every long-term governance option on the table — continued LLC, worker cooperative, ESOP, hybrid — for the partnership to evaluate later when we have operating history. The commodity-factory model would foreclose most of them. Nothing in this memo presupposes a specific legacy outcome.

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- Add capital to the Phase 1 budget. Site footprint and building layout decisions per Section 4 may shift Phase 1 capital requirements modestly (a larger lot costs more; a building shell sized for future expansion costs more), but the magnitude is small relative to Phase 1 total capital and is offset by avoided Phase 2-3 retrofit costs.

6. What I'm Asking From Each of You

Partner	Ask
Trevor (GMD)	Reframe the site search to evaluate Phase 3 ultimate footprint (8–12 acres) rather than only Phase 1 immediate needs (1–6 acres). Re-engage OED with the broader site question. Flag immediately if Riverfront East Parcel B advances — project-specific ZÜBLIN outreach activates with Parcel B.
Leo (Operations)	Add scope-expansion and legacy-flexibility language to the Merge4Ward LLC operating-agreement drafting checklist before the next session with the Delaware entity-formation attorney. Specific items: future product lines beyond CLT/SIP, sister-entity formation under shared ownership, and flexibility for whatever long-term legacy structure the partnership eventually chooses (partner-owned LLC, worker cooperative, ESOP, or hybrid).
Travis (Delivery)	When the Phase 1 facility architectural design begins, build the program around Phase 3 capacity with phased equipment build-out. Foundation depths, ceiling heights, crane bay dimensions, and electrical service to accommodate the larger expansion. Cost differential at design stage is small; cost of retrofitting later is enormous.
Gwen (LEEP DE)	Confirm that the LEEP DE workforce training program's Phase 1 scope remains unaffected by this memo (it does), and that Phase 2-3 platform expansion creates additional training-track opportunities rather than mission drift (cladding finishing, connector installation, curved-CLT fabrication all sit cleanly inside LEEP DE's workforce mission). No immediate scope change requested.
All five	Decide as a partnership whether to formally adopt the ZÜBLIN-model framing as our Phase 2 evolution direction. If yes, the three Section 4 decisions get incorporated into Phase 1 work immediately. If no, this memo files as a strategic option we considered and rejected. Either answer is workable — what is not workable is making the Phase 1 site / building / operating-agreement decisions without resolving the question first.

7. One Last Thing Worth Naming

The October 2025 strategy document was not wrong. It was the right plan for the partnership we were building at that point — five people forming an LLC around the Claymont opportunity, with a workforce-training partner, a Mayor's-office invitation, and a clear product (CLT/SIP panels) we could execute on. The plan got us this far.

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What the ZÜBLIN model adds is a more ambitious horizon. We are not just building a panel factory in Wilmington. We have the chance to build the first vertically-integrated timber-construction project delivery platform on the Mid-Atlantic seaboard — anchored in Wilmington, with manufacturing as one revenue line among several, with a workforce-training program creating genuine career pathways across multiple disciplines, and with partner equity that compounds across decades rather than oscillating with the construction cycle. The integration is the moat. The moat is what makes this a wealth-building partnership for the five of us over twenty years rather than a five-year run before the next downturn. The moat is also what doubles the workforce footprint we put into Wilmington across Phase 2 and Phase 3.

What the legacy of this company eventually becomes — partner-owned LLC, worker cooperative, ESOP, hybrid, something we have not thought of yet — is a question for a partnership conversation we should plan to have in Year 3 or Year 4. It is not a question for now. What this memo asks is that we keep all of those legacy options open by structuring Phase 1 correctly. The platform model leaves every legacy outcome on the table. The commodity-factory model would foreclose most of them before we ever get to choose.

The European firm we are reading is one that did this thirty years ago and grew to be part of one of the largest construction companies on the continent. That is not a guarantee that we can do the same thing in Wilmington. It is evidence that the path exists. The three decisions in Section 4 are how we keep that path open.

Discussion at the next partnership meeting. I will bring the longer source memo (the one that walked through the full ZÜBLIN brochure analysis) to the meeting for anyone who wants to see the underlying material. This summary is the version meant for partnership-level decision-making.

— David

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